Learning-to-Rank

CP4285 Modern Recommendation Systems — Week 08

CP4285 Instruction Team

06 Oct 2026

Scan for the slides
soc-n.us/cp4285-t2610-w08

01 Announcements and Context

CHANGELOG

- **6 Oct 2026** — Added feedback and Canvas Voices; retrofitted activity templates; introduced original LTR papers and a compact McAuley BPR summary; developed varied calculation builds and a timed Kendall’s tau exercise; removed the objective-clinic activity.
- **5 Oct 2026** — Added the next-week prelecture exercise, submission requirements, and lecture preview.
- **24 Aug 2026** — Refreshed the course slide deck and changelog.

Bandit Time: Start TempoQuiz

TEMPOQUIZ HOST LAUNCH

Open the room before Question 1

Open the host dashboard, then ask students to join with the **live QR code or room code shown there**.

LIVE QR + ROOM CODE

Use the host dashboard — do not add a static code here.

HOST DASHBOARD CUE

Week 07 Recap

ROOM
Open
students can join

QUESTION
1
ready to release

▶ Release question 1

Wait for the room to settle, then release the first recap question.

WEEKS 06 + 07 RECAP

Why not use one deep query-product scoring stage?

SINGLE-STAGE DEEP SCORING

Apply joint attention to every candidate product

Input: K_1 products, each with a query-product sequence of length L.

APPROXIMATE ATTENTION COMPUTE

$O(K_1 L^2 d)$

K_1 products × quadratic query-product interaction × hidden dimension d

With $K_1 = 1{,}000$, every request jointly processes 1,000 product descriptions with Maya's query and session. The most expensive model is spent on many weak matches.

STAGED PRODUCT RANKING

Use deep query-product scoring only after cheaper cuts

Generate: $N \rightarrow K_1$ with a lexical, dense, or hybrid candidate index.

Rank: $K_1 \rightarrow K_2$ with lightweight product and session features.

Re-rank: $K_2 \rightarrow K_3$ with joint query-product scoring, $O(K_2 L^2 d)$.

DEEP-SCORING REDUCTION

K_1/K_2

1,000 → 20 means 50× fewer query-product evaluations

Two towers create a shared comparison space

OFFLINE · PRODUCT TOWER
Product title, description, attributes
SKU-LUG-204 → product vector $\mathbf{v}_p$

→

SHARED VECTOR SPACE
Comparable embeddings
nearer vectors = stronger semantic candidate fit

←

ONLINE · QUERY TOWER
Maya's query + current session
→ query vector $\mathbf{v}_q$

Training: historical sessions and observed product interactions teach the towers which query-product pairs belong near each other.

Retrieval: compare $\mathbf{v}_q$ with cached $\mathbf{v}_p$ vectors; do not run the product tower again for every query.

IVF: search selected vector clusters

INVERTED-FILE (IVF) INTUITION

Cluster product vectors before queries arrive

5M vectors offline k-means → **4,096 lists** coarse clusters → **Probe 32 lists** search a fraction

AVERAGE-WORK ESTIMATE

`nprobe` is a recall–latency control

APPROXIMATE VECTORS SCANNED

≈ 39,000

5,000,000 × 32 / 4,096 = 1/128 of a full scan, before coarse assignment

`IndexIVFFlat` keeps full vectors; IVF reduces search work, not raw-vector storage. IVF-PQ can reduce storage at additional approximation cost.

Project Critique: your assigned lens

Give one constructive, specific comment on the criterion assigned to your peer team. Each team has a five-minute hard presentation limit; staff and peers ask questions afterwards.

1 · Presentation

- Clear motivation
- Logical structure and layout
- Appropriate images, plots, or figures

2 · Methodology

- Task, dataset, and related resources
- Proposed approach and evidence so far
- Current challenges or open questions

3 · Administration

- Distribution of workload
- Next steps, milestones, and timeline
- One actionable improvement

Post your assigned peer comment on Canvas by Mon, 5 Oct 2026 · 23:59 SGT.

Week 8: Learning-to-Rank

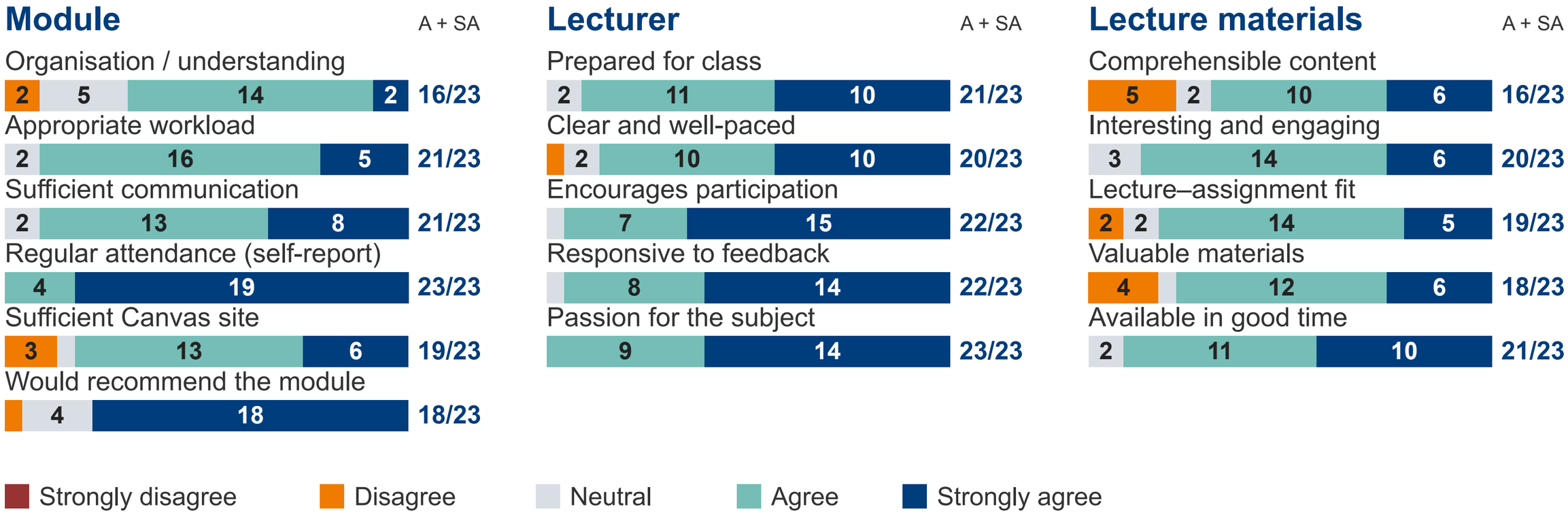
Learning outcomes

- Formulate ranking objectives.
- Compare ranking approaches.
- Analyze position bias.
- Discuss fairness implications of ranking.



Midterm feedback: survey ratings

23 responses per item · Bar segments show counts · A + SA = agree + strongly agree



What should we preserve?

Participation and responsiveness: each has 22/23 agree or strongly agree.

What would make learning clearer?

Content comprehensibility: 5/23 disagree. Materials' value: 4/23 disagree. Which concrete change would help?

CP4285 teaching team. (2026). *Midterm survey* [Course survey]. Canvas. Respondents only; not a claim about the whole class.

Midterm feedback: learning from lectures

Anonymous student excerpts · Theme labels, colour and normal/italic type distinguish clusters

Structure & clarity

“The point being brought across each slide is not clear, nor is the link or connection between the different content.”

“vague sentences that sound legit but don't really help with learning.”

Worked examples & practice

“lecturer can use whiteboard more to explain concepts, especially the mathematical explanations”

“more relevant practice examples so that we can apply the concepts more effectively.”

Technical depth & deployments

“Some concepts taught are still a little high-level for my taste”

“a short skim or walkthrough of more papers / real world deployments such as FAISS”

Peer discussion

“having more opportunities to discuss in-class with other group members around us can help us retain what has been taught in class better.”

Discuss: Which change would help most: clearer sequencing, worked examples, or optional deeper walkthroughs?

CP4285 teaching team. (2026). *Midterm survey* [Course survey]. Canvas. Selected verbatim phrases; theme headings are instructor summaries.



Midterm feedback: assessment & course design

Anonymous student excerpts · Theme labels, colour and normal/italic type distinguish clusters

Quiz timing & question quality

"The midterm quiz timing is confusing (25min actual vs 30min time limit in Canvas)"

"Questions could have been made clearer I feel."

Essay scope & grading criteria

"There could also be clearer rubrics or deliverables to the essays."

"I think the essay was too open ended in general to write a good response"

Participation-mark allocation

"maybe can convert the admin items that weighs 1% to be assigned for the Tempo quizzes."

Team formation

"I think its better to let students form their whole team."

"groupmates do not have the same baseline knowledge of the subject."

Discuss: What can we clarify now—and which proposals need a course-level decision?

CP4285 teaching team. (2026). *Midterm survey* [Course survey]. Canvas. Selected verbatim phrases; theme headings are instructor summaries.

LAST WEEK PRE-LECTURE EXERCISE · W07

One list, two plausible orders

Canvas discussion board · complete by **Mon, 5 Oct 2026 · 23:59 SGT**

> **Choose five items from a recommendation domain you used this week. Order the same set twice for different reasons. Explain the reason by identifying one item whose position changes, and explaining why.**

Your post: Name the domain, list both complete orders, and explain the changed position.

Then post a constructive reply to one classmate's post.
Name and apply one treatment lens in your comment: ♠ ♦ ♥ ♣.



LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

Travel & transport

Travel · cost vs novelty

“If simply travelling is the goal i.e just leave Singapore then Malaysia is the most cost-effective destination to go to.”



POST

“Your two orders are almost exact reverses, so "novelty" ends up just meaning "far and expensive.""”

REPLY · ♠ SPADES



Transport · cost vs effort

“… walking ranks last as it requires the most exertion, especially over longer distances or in hot and hazy weather.”



POST

“… it would be interesting to see how the Diamonds lens can be factored in …”

REPLY · ♦ DIAMONDS QUESTION



TikTok & YouTube feeds

TikTok · engagement vs discovery

“In order B, ranked for diversity and discovery, the ramen video moves up to the top.”



POST

“This can make the FYP feel less repetitive and improve discovery.”

REPLY · ♦ DIAMONDS



YouTube · personal interest vs trends

“If I am trying to relax and watch my favourite YouTuber, then I would like to see Order A.”



POST

“… although relying heavily on trends could also create conformity where users mainly see what is already popular.”

REPLY · ♥ SOCIAL LENS



02 Ranking Objectives



One ranked list, three learning targets

Pointwise

Is this item relevant?

Learn a score or label for each user–item pair independently.

Useful when explicit and calibrated scores are given.

Pairwise

Which of two items comes first?

Learn that an observed or preferred item should outrank a comparison item.

Useful when relative preference is clearer than a rating.

Listwise

How should the candidate group rank?

Define a loss using the candidate group, such as a target distribution.

Position-sensitive metric alignment depends on the training loss and update weighting rules.



Sourcing Internships for Maya

All four roles satisfy the student's hard constraints. Grades are independent assessments of fit, constructed for this example; urgency is held fixed.

Company	Akari	Banyan	Chandra	Dempsey
Current interface position	1	2	3	4 · hidden
Assessed relevance grade y_i	1	2	0	3
Current model score s_i	2.5	2	1.5	1
Alternative model score s_i^{alt}	2.5	3.5	1.5	4.5
Alternative interface position	3	2	4 · hidden	1

Two hypothetical models: alternative scores are the assessed grades plus 1.5. Sorting them gives **Dempsey → Banyan → Akari → Chandra**.

> **Note**
>
> Only the **three highest-scoring** roles are shown. The gray Dempsey column marks its exclusion under the **current model**; the alternative displays Dempsey first and excludes Chandra.

Pointwise: estimate relevance one item at a time

$$\hat{y}_{ui} = f(u, i, x_{ui})$$

Training question

"How relevant is this internship to this student?"

Compare a predicted score with a label, rating, click, or conversion proxy.

Ranking limitation

Good independent scores do not directly say which **swap** in the final list matters most.

A small score error at rank 1 can be more consequential than a larger one far below the fold.



Lower prediction error can give the worse list

As a contrast to ranking quality, assume we measure relevance-prediction error:

$$L_{\text{point}} = \frac{1}{4}\sum_{i \in I}(s_i - y_i)^2$$

Company order: Akari, Banyan, Chandra, Dempsey. Fixed grades $y = [1, 2, 0, 3]$. The current model gives $s = [2.5, 2, 1.5, 1]$; the alternative gives $s^{\text{alt}} = [2.5, 3.5, 1.5, 4.5]$.

$$L_{\text{current}} = \frac{(2.5-1)^2 + (2-2)^2 + (1.5-0)^2 + (1-3)^2}{4} = \frac{2.25 + 0 + 2.25 + 4}{4} = 2.12$$

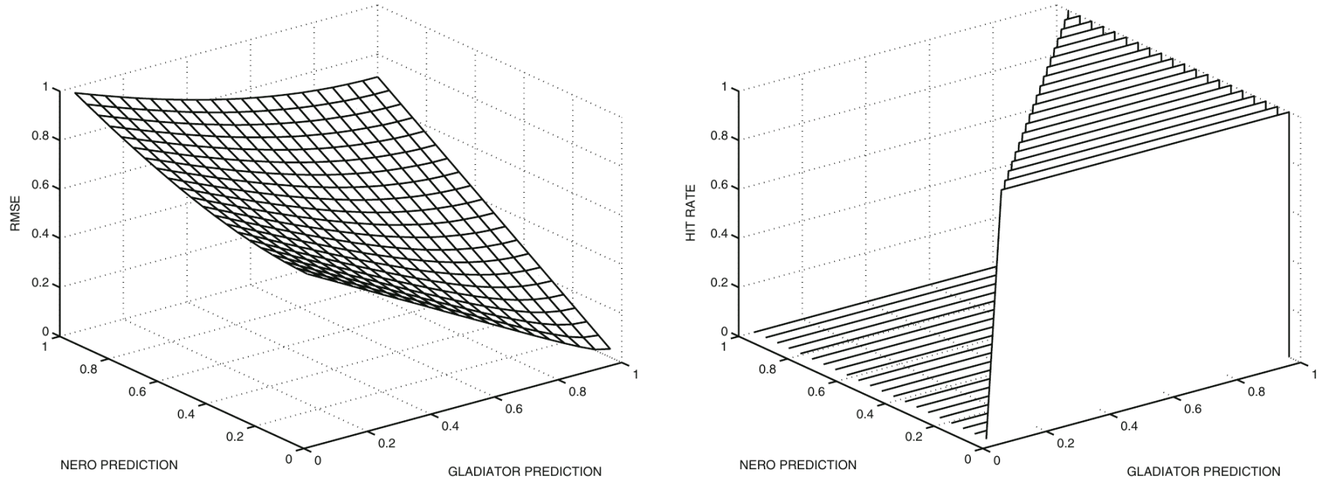
$$L_{\text{alternative}} = \frac{(2.5-1)^2 + (3.5-2)^2 + (1.5-0)^2 + (4.5-3)^2}{4} = \frac{4(1.5^2)}{4} = 2.25$$

Smaller MSE, worse top three: current scores exclude Dempsey. The alternative has a uniform +1.5 score error, yet gives the ideal order **Dempsey, Banyan, Akari, Chandra**.



Why not differentiate the ranking metric?

True ratings: **Nero = 0, Gladiator = 1**. Recommend only the higher-scoring movie.



(a) Smooth RMSE objective (b) Non-smooth hit rate

Figure 13.1: Ranking objective is not a smooth function of predicted ratings (and underlying model parameters)

Source: Charu C. Aggarwal, *Recommender Systems: The Textbook* (Springer, 2016), Figure 13.1, p. 414. Figure reproduced from the textbook; © Springer International Publishing Switzerland 2016.

Flat regions give no useful gradient; the ordering boundary gives a jump. A smooth pairwise surrogate supplies a learning direction without differentiating the discrete metric.

Foundational paper: learning from relative evidence

Joachims (KDD 2002): *Optimizing Search Engines using Clickthrough Data* 🔗.

Training evidence

Derive query-specific document preferences from clickthrough logs.

Clicks supply relative evidence under assumptions about the displayed ranking.

Clicked below a skipped link

Clicking link **3** after passing link **2** suggests **3 should rank above 2**.

Clicking link **7** suggests it should rank above the skipped links **2, 4, 5 and 6**.

Assumes top-to-bottom scanning and informative abstracts. These are partial, potentially noisy preferences—not absolute relevance labels.

ⓘ Why no preference pair from link 1?

Link **1** was clicked, but there are **no skipped links above it**. This rule therefore extracts no pair from that click. It also gives **no ordering between the clicked links 1, 3 and 7**.

From preference pairs to a training metric

You have five inferred preferences: 3 above 2; 7 above 2, 4, 5 and 6. How would you reward a model for respecting them?

Compare two proposed rankings

Ranker X: 1 → 2 → 3 → 4 → 5 → 6 → 7
Ranker Y: 1 → 7 → 3 → 2 → 4 → 5 → 6

Count respected and reversed preference pairs for each ranker. Ignore pairs for which you have no evidence.

Design the metric

Propose a normalized score—or a loss to minimize. State its best and worst values.

05:00

▶ Start Timer

Groups · 5 minutes: write the two scores and your metric definition, including its best and worst values. **Then:** 1 minute to report and discuss.

Hint · Kendall's tau (complete rankings, no ties)

$\tau = (P - Q)/(P + Q)$: **P** counts concordant pairs (same order); **Q** counts discordant pairs (reversed order).

For m items, $P + Q = \binom{m}{2}$; +1 is perfect agreement, −1 complete reversal. Adapt the idea to the five observed pairs; unknown preferences are not errors. This is not full-list tau.

Original RankNet paper: learning to rank by gradient descent

Burges et al. (ICML 2005): *Learning to Rank using Gradient Descent* 🔗.

- Score each query–document feature vector with a shared neural network.
- Turn score differences into pairwise preference probabilities.
- Train with probabilistic pairwise loss and backpropagation.

> **Note**
>
> Training compares **pairs**. Serving still scores each candidate and sorts the scores.



Pairwise: learn the direction of a comparison

Compare items **within the same user or query**.

Desired preference

Grades: Dempsey = 3; Akari = 1.

Dempsey should rank above Akari.

Current model disagrees

Scores: Dempsey = 1; Akari = 2.5.

$$\Delta = s_{\text{Dempsey}} - s_{\text{Akari}} = 1 - 2.5 = -1.5$$

Training should increase this margin.

Training family	Pair evidence	Training question
Pairwise ranking	Grades or explicit preferences	Does Dempsey score above Akari?
BPR	Observed interaction vs sampled unobserved item	Does the observed item score higher?

BPR is a method in the pairwise family. Grades give six strict pairs: Dempsey > Banyan > Akari > Chandra.



Worked pairwise example: Dempsey vs Akari

Return to the **assessed-grade** pair **Dempsey > Akari**. Scores are $s_{\text{Dempsey}} = 1$, $s_{\text{Akari}} = 2.5$:

Calculate: $\Delta = s_{\text{Dempsey}} - s_{\text{Akari}}$, $\sigma(\Delta) = 1/(1 + e^{-\Delta})$, $L = -\ln \sigma(\Delta)$.

$$\Delta = 1 - 2.5 = -1.5, \quad \sigma(-1.5) = \frac{1}{1 + e^{1.5}} = 0.1824, \quad L = -\ln \sigma(-1.5) \approx 1.701$$

$$\frac{\partial L}{\partial \Delta} = \sigma(\Delta) - 1 = -0.8176, \quad \frac{\partial L}{\partial s_{\text{Dempsey}}} = -0.8176, \quad \frac{\partial L}{\partial s_{\text{Akari}}} = +0.8176$$

For **independent score parameters** and $\eta = 0.1$:

$$s'_{\text{Dempsey}} = 1 - 0.1(-0.8176) = 1.0818, \quad s'_{\text{Akari}} = 2.5 - 0.1(+0.8176) = 2.4182$$

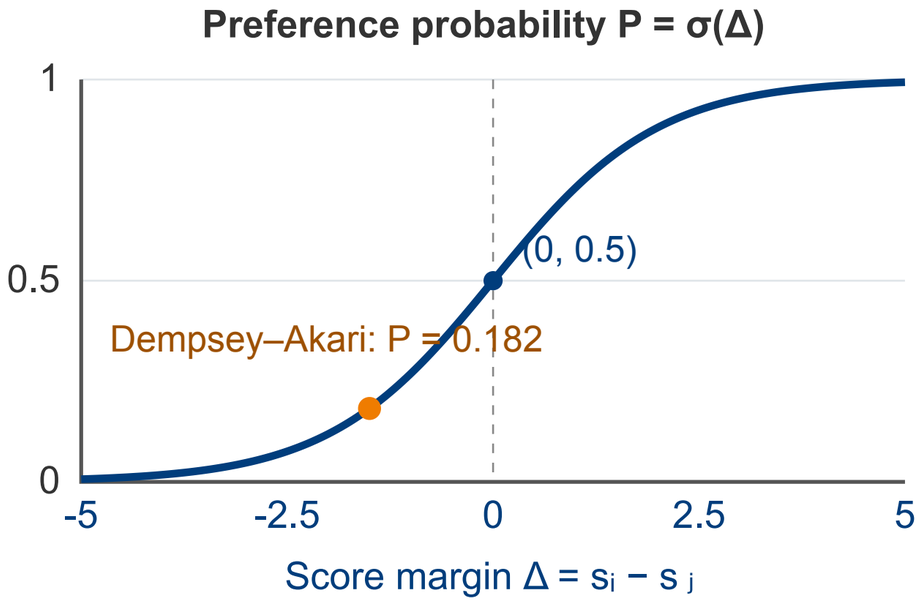
Shared models propagate these signals through their parameters; these are illustrative score updates.



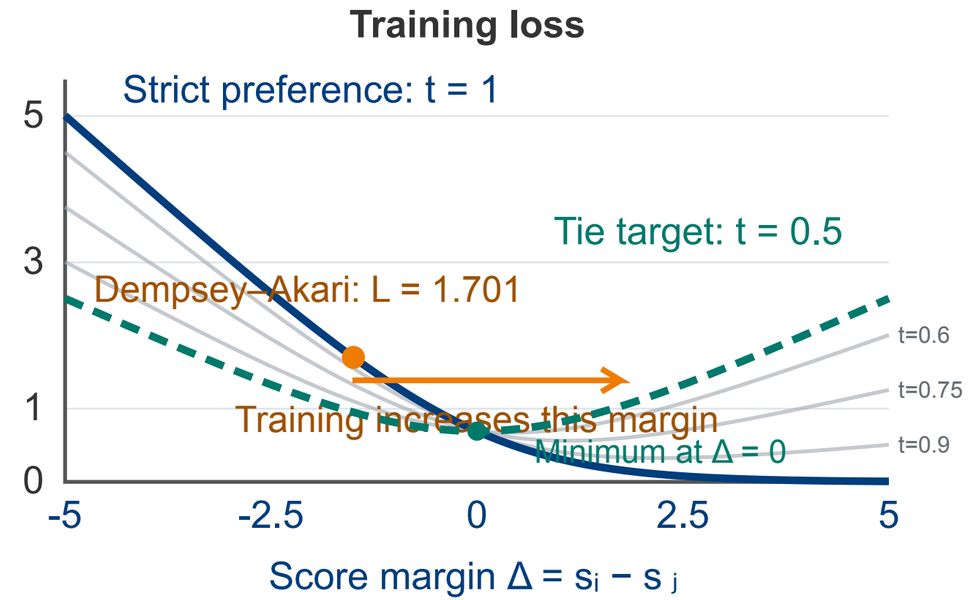
RankNet: a probabilistic pairwise loss

For one query/context, define the score margin $\Delta = s_i - s_j$. Target t_{ij} expresses whether item i should outrank item j.

$$P_{ij} = \sigma(s_i - s_j), \qquad L_{ij} = -t_{ij} \ln P_{ij} - (1 - t_{ij}) \ln(1 - P_{ij})$$



Negative margin: j scores higher · Positive margin: i scores higher



Gray: soft preferences t = 0.6, 0.75, 0.9. Navy: strict preference t = 1.

Tie target (t = 0.5): equal scores minimize loss. Our internship examples use strict preferences (t = 1).



BPR: pairwise ranking from implicit feedback

$$T = \{(u, i, j) : i \in I_u,\ j \notin I_u\}, \qquad \Delta_{uij} = s(u, i) - s(u, j)$$

Here I_u contains the items user u interacted with. **T defines assumed training preferences**, not proven dislikes.

Objective versus model

BPR specifies the training criterion. BPR-MF uses matrix factorisation to supply the score.

Other differentiable scoring models can use the same pairwise loss.

Observed does not mean universally preferred

A sampled unobserved item may be unseen, not disliked. The sampler determines which comparisons training encounters.



BPR: turn preferred orderings into learning updates

McAuley §5.2.2: learn preferred orderings, not absolute ratings.

1 · Define success

$x_{uij} = s(u,i) - s(u,j)$

Positive → **i ranks above j**.

Observed i vs sampled unseen j. AUC counts correct pairs; the hard count has no useful gradient.

2 · Use a smooth loss

$P = \sigma(x_{uij})$: predicted pair preference.

BPR assumes the strict target **t = 1**.

$L_{\text{pair}} = -\ln P = -\ln \sigma(x_{uij})$

Same cross-entropy as S22 with t = 1. Sum pair losses + L2 regularization; minimize L.

3 · Update the model

$\partial L/\partial x = P - t = \sigma(x_{uij}) - 1$

Upward correction: $t - P = 1 - \sigma(x_{uij})$.

With t = 1, lower predicted preference means a stronger correction. The chain rule updates the model parameters.

4 · See the direction

Dempsey > Akari: scores 1 vs 2.5.
Margin −1.5 → probability 0.1824.
Loss 1.701 → push Dempsey up, Akari down.

Grade-based illustration of the loss; BPR uses implicit-feedback pairs.

Break — 5 minutes

WHILE YOU PAUSE · RECOMMENDATION HISTORY

2010 · Yahoo’s ranking challenge

An ensemble of LambdaMART models won Track 1 of the Yahoo! Learning to Rank Challenge. Search ranking made a practical case for learning from errors that matter to the ranked list (Burges et al., 2011).

Why should a swap near rank 1 carry more learning pressure than a swap near rank 40?



▶ Start Timer

Burges, C., Svore, K., Bennett, P., Pastusiak, A., & Wu, Q. (2011). Learning to rank using an ensemble of lambda-gradient models 🔗. In O. Chapelle, Y. Chang, & T.-Y. Liu (Eds.), *Proceedings of the Learning to Rank Challenge* (Vol. 14, pp. 25–35). PMLR.

03 Listwise Ranking & LambdaRank



Listwise quality: measure the displayed order

Relative order, not position weighting: basic BPR has no explicit NDCG swap weights.

1 · Convert grades to gains

$$g_i = 2^{y_i} - 1$$

Company	Grade	Gain computation
Akari	1	$2^1 - 1 = 1$
Banyan	2	$2^2 - 1 = 3$
Chandra	0	$2^0 - 1 = 0$
Dempsey	3	$2^3 - 1 = 7$

Gains come from assessed grades. Dempsey contributes zero to the current DCG@3 because it is at rank 4.

2 · Discount gains by position

$$DCG@3 = \sum_{r=1}^{3} \frac{g_{\pi_r}}{\log_2(r+1)}$$

Rank 1/2/3 discounts: 1, 0.631, 0.5.

Current: Akari → Banyan → Chandra

$$DCG = 1(1) + 3(0.631) + 0(0.5) = 2.893$$

Ideal: Dempsey → Banyan → Akari

$$IDCG = 7(1) + 3(0.631) + 1(0.5) = 9.393$$

3 · Normalize

$$NDCG@3 = \frac{2.893}{9.393} \approx 0.308$$ **· Ideal = 1.**

ListNet: learn from a candidate-group distribution

Cao et al. (ICML 2007): *Learning to Rank: From Pairwise Approach to Listwise Approach* 🔗.

One training example = one candidate group

Fix a query or user/context, then take its candidates and relevance grades together.

For Maya: all four internships form one example. Compute one loss across the group, rather than a separate loss for each item or pair.

“Top-one” gives every item a probability

The model scores every candidate. Softmax turns these scores into a distribution: each value expresses that item's probability of occupying first place under the model.

Sort all probabilities to rank the whole group. This gives exactly the same order as sorting the original scores.

$$p_i = \frac{e^{y_i}}{\sum_j e^{y_j}}, \qquad q_i = \frac{e^{s_i}}{\sum_j e^{s_j}}, \qquad L_{\text{list}} = -\sum_i p_i \log q_i$$

p_i = target top-one probability, derived from item i’s assessed relevance grade. **q_i = predicted top-one probability**, derived from its model score. Each distribution sums to 1 across the candidate group.

Training: bring q closer to p. **Serving:** sort model scores; softmax preserves their order.

ListNet: turn four scores into one group loss

Company order: **Akari, Banyan, Chandra, Dempsey** · Grades $y = [1, 2, 0, 3]$ · Scores $s = [2.5, 2, 1.5, 1]$

1 · Exponentiate, then normalize

$$Z_y = e^1 + e^2 + e^0 + e^3 = 31.193$$

$$Z_s = e^{2.5} + e^2 + e^{1.5} + e^1 = 26.772$$

Divide each exponential by its group total.

Akari: $p_{\text{Akari}} = e^1/31.193 = 0.0871$
Dempsey: $p_{\text{Dempsey}} = e^3/31.193 = 0.6439$

Akari: $q_{\text{Akari}} = e^{2.5}/26.772 = 0.4551$
Dempsey: $q_{\text{Dempsey}} = e^1/26.772 = 0.1015$

2 · Compare the distributions

$$p_{\text{target}} \approx [0.0871,\ 0.2369,\ 0.0321,\ 0.6439]$$

$$q_{\text{predicted}} \approx [0.4551,\ 0.2760,\ 0.1674,\ 0.1015]$$

Each sums to 1. The target favours Dempsey; the model favours Akari.

3 · Cross-entropy (natural logs)

$$L = -\sum_i p_i \ln q_i$$

$$L \approx -[0.0871 \ln(0.4551) + 0.2369 \ln(0.2760) + 0.0321 \ln(0.1674) + 0.6439 \ln(0.1015)] \approx 1.904$$

Wrong order · Dempsey vs Akari
Grades: 3 > 1 · **Scores:** 1 < 2.5
Item gradients (q − p):
Dempsey: $0.1015 - 0.6439 = -0.5424$
Akari: $0.4551 - 0.0871 = +0.3680$
Update: Dempsey ↑; Akari ↓. Large corrections.

Correct order · Banyan vs Chandra
Grades: 2 > 0 · **Scores:** 2 > 1.5
Item gradients (q − p):
Banyan: $0.2760 - 0.2369 = +0.0391$
Chandra: $0.1674 - 0.0321 = +0.1353$
Update: Banyan ↓ slightly; Chandra ↓ more.

LambdaRank: metric-aware pairwise learning

Burges, Ragno and Le (NeurIPS 2006): *Learning to Rank with Nonsmooth Cost Functions* 🔗.

RankNet does not account for NDCG

Its pairwise loss uses score margins, without NDCG's rank discounts, top-K cutoff or swap-impact weights.

Equal margins give equal strict-pair corrections—even when one swap would improve NDCG much more.

LambdaRank adds NDCG sensitivity

Scale each pairwise learning signal by the absolute change in NDCG caused by swapping those items.

With the same margin, a larger NDCG improvement produces a stronger correction.

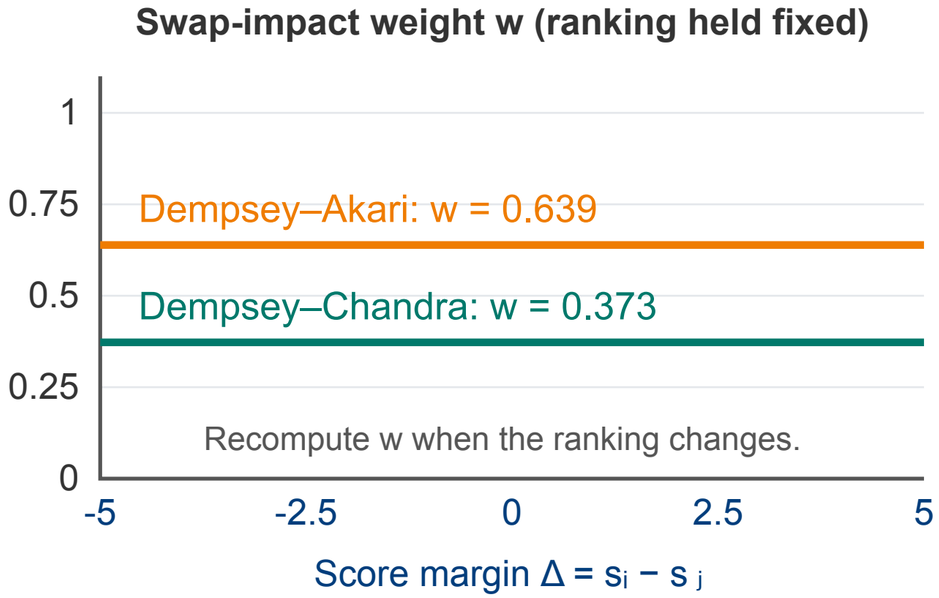
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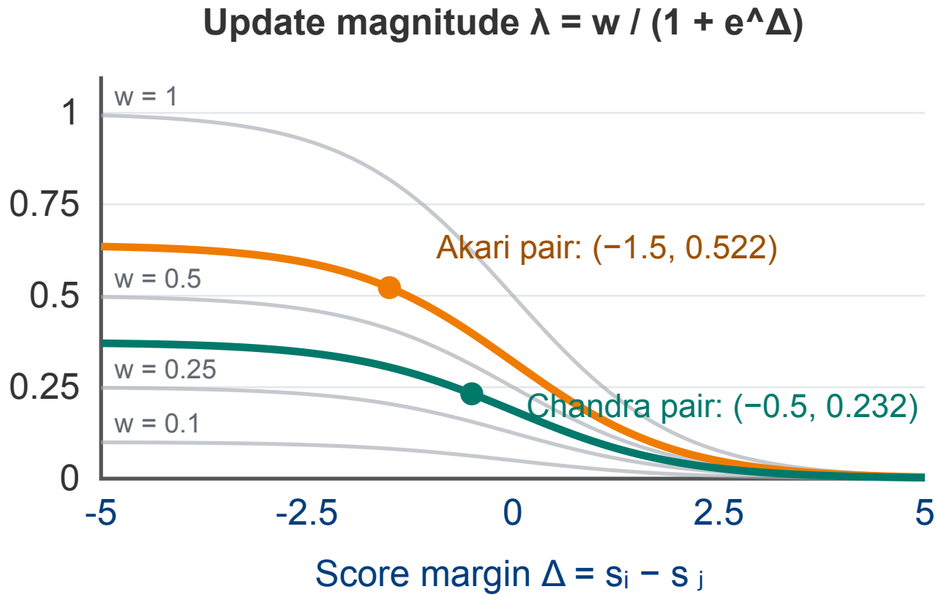
LambdaRank specifies **score-update signals** and propagates them through a scorer. LambdaMART later uses such signals with boosted trees.

LambdaRank: weight a comparison by its list impact

Same RankNet variables: $\Delta = s_i - s_j$, $P_{ij} = \sigma(\Delta)$, strict target $t_{ij} = 1$. New weight: $w_{ij} = |\Delta NDCG@3_{ij}|$.

$$\lambda_{ij} = w_{ij}(t_{ij} - P_{ij}) = w_{ij}(1 - P_{ij}) = \frac{w_{ij}}{1 + e^{\Delta}} \qquad (t_{ij} = 1)$$





Swap impact scales the curve; margin selects the point. Positive λ pushes the preferred item's score up and the other down; sum pair signals through the model.

Two swaps: work out their top-three utility

Current order **Akari, Banyan, Chandra, Dempsey**; gains **1, 3, 0, 7**. Current DCG = **2.893**, IDCG = **9.393**. Beyond rank 3, the discount is **zero**.

1 · Dempsey ↔ Akari: new top three Dempsey, Banyan, Chandra.

$$DCG' = 7(1) + 3(0.631) + 0(0.5) \approx 8.893, \quad NDCG' \approx 8.893/9.393 = 0.947$$

$$w_{\text{Dempsey,Akari}} \approx |0.947 - 0.308| = 0.639$$

2 · Dempsey ↔ Chandra: new top three Akari, Banyan, Dempsey.

$$DCG' = 1(1) + 3(0.631) + 7(0.5) \approx 6.393, \quad NDCG' \approx 6.393/9.393 = 0.681$$

$$w_{\text{Dempsey,Chandra}} \approx |0.681 - 0.308| = 0.373$$



BPR in the learning-to-rank map

Method	Training signal	Correction
Pointwise	One item and its label	Item-level prediction error
RankNet	Relevance preference pair	Logistic pairwise error
BPR	Observed / unobserved user pair	Logistic error + pair sampling
LambdaRank	Pair in the current ranking	Pairwise signal × swap impact
ListNet	Candidate-group distribution	Group cross-entropy

BPR-MF = a matrix-factorisation scorer trained with BPR. Neither a neural scorer nor BPR alone specifies fair exposure.



04 Position Bias & Fair Ranking



Click counts need exposure context

A job card gets **60 clicks at rank 1** and **18 clicks at rank 6**.

Without impression counts or evidence about examination, we cannot tell whether this reflects relevance or simply more opportunity to be seen.

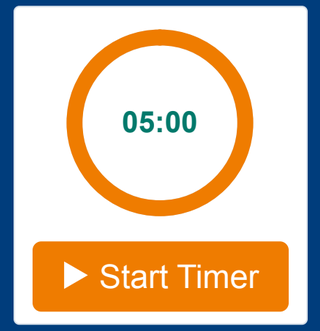


👥 Evidence audit: what can you conclude?

Same job card, comparable audiences. Data are constructed; examination probabilities are supplied assumptions.

Rank	Impressions	Clicks	Assumed P(E)
1	1,000	60	0.60
6	1,000	18	0.18

Raw CTR = clicks / impressions
Adjusted estimate = raw CTR / assumed P(E)



Groups · 5 minutes · one deliverable: write a two-sentence verdict on the claim **"The card is more relevant at rank 1."** Support your verdict with the exposure-adjusted estimates and one limitation of that evidence.

Position bias: clicks mix relevance with opportunity

Clicks combine examination opportunity and response after examination.

$$P(C = 1 \mid i, r) = P(E = 1 \mid r)\, P(C = 1 \mid E = 1, i)$$

Model assumptions: clicks require examination; examination depends on rank; response after examination is rank-invariant. Clicks remain a proxy for relevance.

Log exposure

Record impressions, rank, and context. Unseen is not rejected.

Test carefully

Use randomisation or defensible propensity-aware methods.

Audit allocation

Check who receives exposure; specify a fairness guardrail.

Summary



🔑 A ranking objective encodes which errors matter most.

1 · Decide what counts as evidence.
Clicks can supply partial, noisy preference pairs; relevance grades express an assessed ordering. Declare the assumptions behind either signal.

2 · Choose what training should correct.
Pointwise reduces item prediction error; RankNet learns pair preferences; BPR constructs strict pairs from implicit feedback; ListNet matches candidate-group probability distributions.

3 · Make position matter when the goal requires it.
Lower prediction error can still give a worse top-three list. RankNet has no explicit NDCG weighting; LambdaRank scales pair corrections by NDCG swap impact.

4 · Separate relevance from opportunity.
Exposure changes clicks. Adjusted estimates depend on examination assumptions; equal estimates alone do not establish fairness.

Next week: use graph connections to explore evidence beyond an item’s interaction count.



NEXT WEEK PRE-LECTURE EXERCISE

Popularity or connection? Fix the surface

Canvas · Solo · 15 minutes · no peer reply

Complete by Mon, 12 Oct 2026 · 23:59 SGT

Choose **one surface** and keep its user, task, and constraints fixed throughout your answer.

A · Music autoplay

Next track after an acoustic-folk playlist. Keep the genre transition gentle.

B · Jobs search

First search-result card for a graduate seeking entry-level data work in Singapore.

C · Grocery checkout

One “Complete your meal” checkout item for a vegetarian dinner basket. It must be available.

Both candidates are eligible: X has 1,000 interactions but no known overlap with this user’s interests; Y has 20 interactions, several from users with overlapping histories. Counts come from the same period and interaction definition.

Post 120–160 words: name your surface and concrete X/Y items; choose which to investigate first; state one assumption, one check of exposure, and evidence that would reverse your choice.

Next lecture: ask what the pattern of connections reveals beyond an item’s total count.



NEXT WEEK PREVIEW — WEEK 09

Graph-Based Recommendation

Image generated with GPT